

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

1898

Paper Title

A Structured Approach to Insurance Claims Data Processing: From Manual Extraction to Automated Matching

Abstract

Insurance datasets often contain inconsistencies, formatting issues, and unstructured information, making data processing and matching difficult. This research presents a structured and scalable approach for improving insurance claims data processing through a step-by-step methodology involving manual extraction, data standardization, and automated matching.

In the first stage, important fields such as loss location, date of loss, insured details, and claim values were manually extracted from semi-structured insurance reports to understand real-world data challenges. In the second stage, Power Query and SQL were used to clean, standardize, and consolidate datasets by handling issues related to text formatting, currency values, date inconsistencies, and location ambiguities. The final stage introduced an automated claims matching and scoring system that linked claims data with exposure datasets using tokenization, fuzzy matching, Jaccard similarity, and weighted scoring logic.

The results showed significant improvements in processing speed, consistency, and matching accuracy. The study also highlighted the importance of continuous data cleaning, incremental development, and simple yet reliable logic in building effective data systems. Furthermore, the use of modular transformation functions and audit mechanisms improved the maintainability, scalability, and transparency of the system. Overall, this research demonstrates how combining manual understanding with automation technologies such as Power Query, SQL, Python, and NLP can create a more reliable and efficient insurance data processing framework.

Created

2026-05-11 22:27:23 GMT+5:30

Last Modified

2026-05-11 22:27:23 GMT+5:30

Authors

Mayank Mayank (Chitkara University) <mayank565.be22@chitkara.edu.in>

Submission Files

Mayank_Correctedfile_25-04-2026_tobefinal.pdf (372.6 Kb, 2026-05-11 22:25:07 GMT+5:30)

Supplementary Files

Mayank_Correctedfile_25-04-2026_tobefinal (1) (2).pdf (47.9 Kb, 2026-05-11 22:39:47 GMT+5:30)

Submission Questions Response

1. Corresponding author Email ID

Provide Email ID of corresponding author

mayank565.be22@chitkara.edu.in

2. Contact number with country code

Provide Whatsapp number of corresponding author with country code

+918968036659

3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

Agreement accepted

4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

Agreement accepted

5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

Agreement accepted

6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

Agreement accepted
